Security Operations Centre
Incident Investigation Report

Source: Scenario_2_Alerts.json and Scenario_2_Archives.json

Endpoint: DESKTOP-5OG7C28 (Wazuh agent 001, Windows_VM, 192.168.232.130) | **Date:** 27 October 2026

Event window: 11:29:36.283-11:34:42.648 UTC (12:29:36-12:34:42 +01:00) | **Overall confidence:** MEDIUM

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1. Executive Summary

A high-integrity Windows PowerShell process on a single workstation launched rapid host/user identification commands and a **cmd.exe** chain performing local account discovery, local-group enumeration, directory-listing instructions and Windows Credential Manager enumeration. The core activity comprises **14 Sysmon process-creation events in approximately 0.931 seconds** (Sysmon records 28312-28325), all attributed to **DESKTOP-5OG7C28\Darragh Casey**, logon ID **0xFCD15**.

The credential-access component is independently corroborated: Security event **5379 / record 83505** reports a successful credential-enumeration operation by **cmdkey.exe PID 7940** with **countOfCredentialsReturned=1** and **returnCode=0**. Security event **4799 / record 83506** records enumeration of the built-in **Users** group by the matching **net1.exe PID 10484**. These are behavioral findings, not merely alert labels.

The sequence is **suspicious for reconnaissance and credential-store access** if not independently authorized. It does not demonstrate credential secret disclosure, outbound transfer, persistence, an initial compromise, or privilege escalation. The earlier opening of a **Test_2.ps1** file in Notepad is contextual but cannot be linked causally to the later process chain from the supplied records.

2. Scope of Activity

Evidence consists of **23 unique Windows events** in Scenario_2_Archives.json: 17 Sysmon Event ID 1 (process creation), two Sysmon Event ID 13 (registry value set), and four Windows Security events (4624, 4672, 5379 and 4799). Scenario_2_Alerts.json contains **eight alerts**, corresponding exactly to Sysmon process records **28318-28325** also present in the archive. These are duplicate representations, **not 31 separate events**.

All records identify one endpoint, **DESKTOP-5OG7C28**. The Wazuh agent metadata supplies IP **192.168.232.130**; it is not a network-connection observation. The decisive process activity belongs to an interactive session (terminal session 1) under **DESKTOP-5OG7C28\Darragh Casey** with **High** integrity and logon ID **0xFCD15**. A separate Notepad event under the same username uses logon ID **0xFCD57** and **Medium** integrity.

The input is a bounded telemetry extract, not evidence of complete host coverage. It includes no command output, executable or script contents, network connection telemetry, file-copy records, or subsequent activity establishing exploitation or exfiltration. The isolated laboratory context affects exposure and attribution but does not make the observed discovery and credential-enumeration behavior intrinsically benign.

3. Detailed Investigation

3.1 Earlier file viewing and unrelated background activity

At **11:29:36.283 UTC** Notepad was launched from Explorer with the argument **C:\Users\Darragh Casey\Desktop\S_2\Test_2.ps1** (Sysmon **28307**). This shows the file was passed to a text editor; it does **not** establish that its PowerShell content executed. The later PowerShell parent is **PID 12276** in logon **0xFCD15**, while Notepad has a different parent GUID, logon ID **0xFCD57**, and Medium integrity. There is no file-content evidence or process-lineage connection between them.

Two **svchost.exe** Sysmon registry writes to a **Compatibility Assistant\Store** key naming Notepad (records **28308-28309**) appear consistent with application-compatibility tracking; they do not show a persistence modification. A SYSTEM service host, a service logon with privileged token (Security **83503-83504**, logon type 5 and logon ID **0x3E7**), and SYSTEM **taskhostw.exe** (Sysmon **28310-28311**) have neither the discovery session logon ID nor an observed parent/child relationship to the later chain. They are excluded from attribution to the primary activity.

3.2 Parent PowerShell and rapid identification commands

Between **11:34:41.717** and **11:34:42.170 UTC**, the same parent PowerShell instance (**PID 12276**; ProcessGuid **{4b2ce92d-8a8c-6ae0-6206-000000001a00}**) generated **four HOSTNAME.EXE** children (Sysmon **28312, 28313, 28314, 28316**) and **two whoami.exe** children (**28315, 28317**). All have the same high-integrity user and logon ID. These commands are directly observed and are consistent with discovering the local computer name and execution identity. Repetition within a fraction of a second suggests a compact or automated workflow, but the script or triggering action is not visible.

There is **no Sysmon process-creation record for powershell.exe PID 12276** in the supplied sample. Its presence and parentage are established from the child process records, not from an observed invocation, script block, or full PowerShell command line.

3.3 Shell command chain and local account/group discovery

At **11:34:42.466 UTC**, PowerShell spawned high-integrity **cmd.exe PID 1832** (Sysmon **28318**; ProcessGuid **{4b2ce92d-8c52-6ae0-8c06-000000001a00}**). The exact command line recorded in the Sysmon system message was:

```
"cmd.exe" /c net user & dir c:\Users\ & cmdkey.exe /list & net localgroup "Users" & net localgroup
```

The five ampersand-delimited commands request: local account listing (**net user**); a directory listing of **C:\Users** (**dir**); Windows Credential Manager listing (**cmdkey.exe /list**); membership of the local **Users** group (**net localgroup "Users"**); and a list of local groups (**net localgroup**). The **dir** instruction is directly present in the shell command line, but the archive contains no console output or separate process record confirming the directory-listing result.

The parent/child ProcessGuid links, rather than timing alone, establish the following three net command pairs. **net.exe** spawns **net1.exe** for each action; their common account, logon ID and High integrity reinforce the association.

Requested operation	cmd.exe child	net.exe child
net user	net.exe PID 12068; S/28319	net1.exe PID 6584; S/28320
net localgroup "Users"	net.exe PID 8816; S/28322	net1.exe PID 10484; S/28323
net localgroup	net.exe PID 6504; S/28324	net1.exe PID 4912; S/28325

Security event **4799 / W/83506** at **11:34:42.612 UTC** confirms that membership of **BUILTIN\Users** (SID **S-1-5-32-545**) was enumerated. The caller process ID **0x28F4** is decimal **10484**, exactly the PID of **net1.exe** in Sysmon **28323**, and the subject logon ID is **0xFCD15**. This establishes the specific command-to-audit relationship. No group modification is observed.

3.4 Credential Manager access

Sysmon **28321** records **cmdkey.exe PID 7940** launched as a direct child of **cmd.exe PID 1832**, with **cmdkey.exe /list** and logon ID **0xFCD15**. Security event **5379 / W/83505** records **Enumerate Credentials** for the same account, **clientProcessId=7940**, process-creation time **11:34:42.5563526Z**, **countOfCredentialsReturned=1**, and **returnCode=0**. The process identity and creation time independently anchor the audit event to the cmdkey process.

This supports a successful enumeration/read operation returning one stored-credential entry. It **does not reveal the credential, prove that a plaintext password was obtained, or demonstrate subsequent use or exfiltration**. It is more significant than a rule description saying an unusual shell executed because it is corroborated by an operating-system audit record.

3.5 Wazuh alert interpretation and supported boundaries

The eight unique alert rows reflect four rule IDs: **92052** (one cmd.exe, level 4), **92032** (one cmdkey.exe, level 3), **92036** (three net.exe, level 3), and **92031** (three net1.exe, level 3). Alerts are useful locators but do not substitute for their underlying events.

In particular, **92036** includes the MITRE **T1574.001** label for DLL search-order hijacking. The supplied evidence instead shows **C:\Windows\System32\net.exe** launching **C:\Windows\System32\net1.exe** for explicitly recorded discovery commands. There is **no supplied DLL load, search-path manipulation or hijacking evidence**; T1574.001 and its associated persistence/evasion/escalation tactics are **not adopted**. Rule **92031** labels all net1.exe events as Account Discovery, while two clearly concern **local group** discovery; rule **92032** tags cmdkey with Account Discovery despite the more specific credential-store action. The report maps the actual commands, not those generic labels.

4. Evidence Analysis

Source of truth: use the embedded **data.win.system.systemTime** UTC and Sysmon **eventRecordID** for sequencing. Windows Security uses its own record-number series. The Sysmon **system.message** field preserves readable command text; some normalized **eventdata.commandLine** fields HTML-escape ampersands as **&**. Readable command quoting above is taken from the system message.

Evidence	Fields / directly observed value	What it supports
S/28312-28317	**ParentProcessGuid** = {4b2ce92d-8a8c-6ae0-6206-000000001a00}; parent PID 12276; four HOSTNAME.EXE and two whoami.exe; High; 0xFCD15	Single PowerShell parent, host and identity discovery.
S/28318	**PID** 1832; parent PID 12276; **CurrentDirectory** = C:\Users\DARRAG~1\AppData\Local\Temp\; command contains net, dir, cmdkey, localgroup	Shell launch, intended sequence; temp current directory is context, not evidence of payload location.
S/28319-28325	**ParentProcessGuid** chains 28318 -> 28319 -> 28320; 28318 -> 28321; 28318 -> 28322 -> 28323; 28318 -> 28324 -> 28325	Exact process relationships for account, credential and group commands.
W/83505	Security EID 5379; **clientProcessId=7940**; **countOfCredentialsReturned=1**; **returnCode=0**; logon 0xFCD15	Successful enumeration of one Credential Manager entry; not password disclosure.
W/83506	Security EID 4799; **callerProcessId=0x28F4** (10484); group BUILTIN\Users; SID S-1-5-32-545	Group-membership enumeration by net1.exe recorded in S/28323.
S/28307-28311; W/83503-83504	Notepad opened a .ps1 in 0xFCD57; SYSTEM background events use logon 0x3E7	Context without direct chain to later 0xFCD15 discovery activity.

Executable identifiers in Sysmon eventdata.hashes (SHA-256): cmd.exe S/28318: 15CD209C995B7D3009C91CA40892D5B8F1FB162F3E5F10716642EA278E8E0D7B; **cmdkey.exe** S/28321: 4125CD7966810CDFAFB18157C276A88E6AC443C7C98D42593C70247045815028; **net.exe** S/28319: BB3E638C8B5B6EF80847E364AEEF2796CAD25D3539CF9B41D3820FA48943E777; **net1.exe** S/28320: 8489DB35375D890314EB8F7CD9E0CBD3866475F4E252865F00F649B0A02FD757. These hashes serve reproducible identification only; they have not been externally checked and do not establish malicious binaries.

Correlation check: the eight alerts match the eight archive records S/28318-S/28325 exactly on their parsed **data** values. A **rule.firedtimes** counter and repeated Wazuh IDs are not counted as additional Windows events. The ordered findings derive from unique Windows event records.

5. Timeline of Events

All dates are **27 October 2026**; timestamps below are **UTC** from the Windows source events, not the Wazuh collection time (which is shown in +01:00). Interleaved Sysmon and Security events have been ordered using their source timestamps.

UTC time	Source records	Event / significance
11:29:36.283	S/28307	Explorer launches Notepad with C:\Users\Darragh Casey\Desktop\S_2\Test_2.ps1; different logon (0xFCD57).

UTC time	Source records	Event / significance
11:29:38.500-11:29:59.615	S/28308-28309	SYSTEM svchost writes Notepad Compatibility Assistant Store value; unrelated to the later command process tree.
11:30:08.786-11:30:08.794	W/83503-83504; S/28310	SYSTEM service logon/privileges (type 5; 0x3E7) and GPSvcGroup svchost start.
11:31:51.233	S/28311	SYSTEM taskhostw.exe created, no relevant common parent or session.
11:34:41.717-11:34:42.170	S/28312-28317	PowerShell PID 12276 launches four HOSTNAME.EXE and two whoami.exe under elevated logon 0xFCD15.
11:34:42.466	S/28318	PowerShell spawns cmd.exe PID 1832 with five-part chained discovery/credential-listing command.
11:34:42.516-11:34:42.533	S/28319-28320	cmd.exe -> net.exe PID 12068 -> net1.exe PID 6584; net user.
11:34:42.561	S/28321	cmd.exe -> cmdkey.exe PID 7940, /list.
11:34:42.570	W/83505	Security 5379: cmdkey PID 7940 enumerated credentials; one entry returned, success code 0.
11:34:42.581-11:34:42.598	S/28322-28323	cmd.exe -> net.exe PID 8816 -> net1.exe PID 10484; localgroup "Users".
11:34:42.612	W/83506	Security 4799: net1.exe PID 10484 enumerates BUILTIN\Users membership.
11:34:42.629-11:34:42.648	S/28324-28325	cmd.exe -> net.exe PID 6504 -> net1.exe PID 4912; localgroup listing.

6. MITRE ATT&CK Mapping

These mappings describe the observed behavior and command intent; they do not assert attacker control or a completed downstream objective. ATT&CK labels supplied in alerts were independently assessed against the records.

Technique	Tactic / name	Telemetry support / limit
T1059.003	Execution - Windows Command Shell	**Observed.** S/28318 cmd.exe /c with a command chain; S/28319-S/28325 form direct process descendants.
T1082	Discovery - System Information Discovery	**Observed.** Four HOSTNAME.EXE executions S/28312-28314 and S/28316.
T1033	Discovery - System Owner/User Discovery	**Observed.** Two whoami.exe executions S/28315 and S/28317.
T1087.001	Discovery - Account Discovery: Local Account	**Observed.** net user via S/28318, S/28319-28320; no /domain or remote scope recorded.
T1069.001	Discovery - Permission Groups Discovery: Local Groups	**Observed.** localgroup "Users" and localgroup S/28322-28325; W/83506 independently verifies Users group-membership enumeration.
T1083	Discovery - File and Directory Discovery	**Command intent only.** S/28318 invokes dir C:\Users\; no output or separate execution audit for the built-in command.
T1555.004	Credential Access - Credentials from Password Stores: Windows Credential Manager	**Observed enumeration.** S/28321 cmdkey.exe /list plus W/83505 successful credential enumeration, count 1. No observed plaintext disclosure.

Mappings not supported: despite the Wazuh 92036 metadata, **T1574.001 DLL Search Order Hijacking** is not supported by these records. No observed token change or new elevation path establishes privilege escalation; High integrity describes the existing process context. A PowerShell parent is identifiable, but there is insufficient PowerShell invocation/script content to characterize a separate PowerShell execution technique beyond the observed parent-child context. No supplied records establish lateral movement, persistence, credential dumping, external communications or exfiltration.

7. Overall Investigation Conclusion

Observed facts (high confidence): On one Windows endpoint, an existing high-integrity PowerShell process in user logon **0xFCD15** launched repeated hostname/identity commands and a chained cmd.exe process. The resulting **net/net1** descendants performed local account and group inquiries. The shell launched **cmdkey /list**; a matching Security 5379 record confirms a successful Credential Manager enumeration returning one entry. Security 4799 verifies local Users-group membership enumeration by the correlated net1 process. The eight Wazuh alerts are duplicates of part of that same telemetry.

Assessment / inference (medium confidence): The rapid elevated sequence is best characterized as **host and account reconnaissance with credential-store enumeration**. It is potentially preparatory activity for unauthorized access, but the data do not distinguish authorized lab operation from malicious operator activity, show how the PowerShell parent was started, or establish that the earlier viewed .ps1 file caused the activity. Neither recovered secrets nor downstream misuse is demonstrated.

Overall confidence: MEDIUM. Confidence in the concrete commands, process relationships and two Security audit corroborations is **HIGH**; confidence in hostile intent, compromise status, execution origin and impact is **LOW to MEDIUM**. The overall rating therefore reflects the unresolved cause and intent, not ambiguity about the logged actions. In an environment without a corroborated authorized test, the behavior warrants a security investigation rather than dismissal on the basis of the lab-like filename.

Appendix A - Source Record Index

The index covers all 23 distinct Windows events in Scenario_2_Archives.json; closely related background and repeat-identification records are grouped. **S/** = Microsoft-Windows-Sysmon/Operational record ID; **W/** = Microsoft-Windows-Security-Auditing record ID. The alert column identifies duplicate alert records present in Scenario_2_Alerts.json. The index is ordered by Windows source UTC time.

UTC	Record / event ID	Primary source observation	Alert rule
11:29:36.283	S/28307 (EID 1)	Notepad.exe [8532]; file Test_2.ps1	-
11:29:38.500-11:29:59.615	S/28308-28309 (EID 13)	Two svchost.exe registry SetValue events for Notepad Compatibility Assistant Store	-
11:30:08.786-11:30:08.794	W/83503-83504 (EID 4624, 4672); S/28310 (EID 1)	SYSTEM service logon and privileges, 0x3E7; GPSvcGroup svchost PID 7360	-
11:31:51.232	S/28311 (EID 1)	taskhostw.exe [PID 11976]; taskhostw.exe	-
11:34:41.717-11:34:42.170	S/28312-28317 (EID 1)	PowerShell PID 12276 spawns HOSTNAME.EXE x4 (PIDs 8624, 6592, 5472, 4068) and whoami.exe x2 (PIDs 11740, 12012)	-
11:34:42.466	S/28318 (EID 1)	cmd.exe [PID 1832]; "cmd.exe" /c net user & dir c:\Users\ & cmdkey.exe /list & net localgroup "Users" & net localgroup	92052
11:34:42.515	S/28319 (EID 1)	net.exe [PID 12068]; net user	92036
11:34:42.533	S/28320 (EID 1)	net1.exe [PID 6584]; C:\WINDOWS\system32\net1 user	92031
11:34:42.561	S/28321 (EID 1)	cmdkey.exe [PID 7940]; cmdkey.exe /list	92032
11:34:42.569	W/83505 (EID 5379)	Enumerate Credentials; client PID 7940; 1 returned; returnCode 0	-
11:34:42.580	S/28322 (EID 1)	net.exe [PID 8816]; net localgroup "Users"	92036
11:34:42.598	S/28323 (EID 1)	net1.exe [PID 10484]; C:\WINDOWS\system32\net1 localgroup "Users"	92031
11:34:42.612	W/83506 (EID 4799)	BUILTIN\Users enumerated; caller PID 0x28F4 (10484)	-
11:34:42.628	S/28324 (EID 1)	net.exe [PID 6504]; net localgroup	92036
11:34:42.648	S/28325 (EID 1)	net1.exe [PID 4912]; C:\WINDOWS\system32\net1 localgroup	92031

Audit method: fields were read from both JSON-line files without external scenario references. Exact alert/archive duplicates were verified by Windows provider and record ID and compared on the parsed data object. Sequence derives from source event time; relationships derive primarily from process GUID, parent GUID, PID and Windows Security client/caller PID, plus account/logon ID. The archive is not assumed to be an exhaustive evidence source.